



# IEC 60870-5-104

## Intrusion Detection Dataset

### Readme File

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## 1. Introduction

The evolution of the Industrial Internet of Things (IIoT) introduces several benefits, such as real-time monitoring, pervasive control and self-healing. However, despite the valuable services, security and privacy issues still remain given the presence of legacy and insecure communication protocols like IEC 60870-5-104. IEC 60870-5-104 is an industrial protocol widely applied in critical infrastructures, such as the smart electrical grid and industrial healthcare systems. The IEC 60870-5-104 Intrusion Detection Dataset was implemented in the context of the research paper entitled "Modeling, Detecting, and Mitigating Threats Against Industrial Healthcare Systems: A Combined Software Defined Networking and Reinforcement Learning Approach" [1], in the context of two H2020 projects: ELECTRON: rEsilient and self-healed EleCTRical pOwer Nanogrid (101021936) and SDN-microSENSE: SDN - microgrid reSilient Electrical eNergy SystEm (833955). This dataset includes labelled Transmission Control Protocol (TCP)/Internet Protocol (IP) network flow statistics (Common-Separated Values (CSV) format) and IEC 60870-5-104 flow statistics (CSV format) related to twelve IEC 60870-5-104 cyberattacks. In particular, the cyberattacks are related to unauthorised commands and Denial of Service (DoS) activities against IEC 60870-5-104. Moreover, the relevant Packet Capture (PCAP) files are available. The dataset can be utilised for Artificial Intelligence (AI)-based Intrusion Detection Systems (IDS), taking full advantage of Machine Learning (ML) and Deep Learning (DL).

## 2. Instructions

The IEC 60870-5-104 dataset was implemented following the methodology of A. Gharib et al. in [2], including **eleven features**: (a) Complete Network Configuration, (b) Complete Traffic, (c) Labelled Dataset, (d) Complete Interaction, (e) Complete Capture, (f) Available Protocols, (g) Attack Diversity, (h) Heterogeneity, (i) Feature Set and (j) Metadata.

A network topology consisting of (a) seven industrial entities, (b) one Human Machine Interfaces (HMI) and (c) three cyberattackers was used to construct the IEC 60870-5-104 Intrusion Detection Dataset. The industrial entities use IEC TestServer<sup>1</sup>, while the HMI uses Qtester104<sup>2</sup>. On the other hand, the cyberattackers use Kali Linux<sup>3</sup> equipped with Metasploit<sup>4</sup>, OpenMUC j60870<sup>5</sup> and Ettercap<sup>6</sup>. The cyberattacks were performed during the following days.

- On Saturday, April 25, 2020, a DoS cyberattack (M\_SP\_NA\_1\_DoS) was executed for 2 hours, using the M\_SP\_NA\_1 command.
- On Sunday, April 26, 2020, two cyberattacks were executed, namely (a) DoS (C\_CI\_NA\_1\_DoS) and (b) unauthorised injection (C\_CI\_NA\_1), using the C\_CI\_NA\_1 command for 2 hours.
- On Monday, April 27, 2020, one unauthorised injection attack (C\_SE\_NA\_1) was executed for 4 hours, using the C\_SE\_NA\_1 command.
- Tuesday, April 28, 2020 two cyberattacks were executed, namely (a) unauthorised injection (C\_SC\_NA\_1) and (b) DoS (C\_SE\_NA\_1\_DoS), using the C\_SC\_NA\_1 and C\_SE\_NA\_1 commands for 2 hours and 4 hours, respectively.
- Wednesday, April 29, 2020, one DoS (C\_SC\_NA\_1) cyberattack was performed for 2 hours, using the C\_SC\_NA\_1 command.
- Friday, June 05, 2020, two cyberattacks were executed, namely (a) DoS (C\_RD\_NA\_1\_DoS) and (b) unauthorised injection (C\_RD\_NA\_1), using the C\_RD\_NA\_1 command for 2 and 4 hours, respectively.

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<sup>1</sup> IEC TestServer - <https://sourceforge.net/projects/iecserver/>

<sup>2</sup> QTester104 - <https://sourceforge.net/projects/qttester104/>

<sup>3</sup> Kali Linux - <https://www.kali.org/>

<sup>4</sup> Metasploit - <https://www.metasploit.com/>

<sup>5</sup> OpenMUC j60870 - <https://www.openmuc.org/iec-60870-5-104/>

<sup>6</sup> Ettercap - <https://www.ettercap-project.org/>

- Saturday, June 06, 2020, two cyberattacks were executed, namely (a) DoS (C\_RP\_NA\_1\_DoS) and (b) unauthorised injection (C\_RP\_NA\_1), using the C\_RP\_NA\_1 command for 2 and 4 hours, respectively.
- Monday, June 08, 2020, a Man In The Middle (MITM) cyberattack was executed for 2 hours, filtering and dropping the IEC 60870-5-104 packets.

For each attack, a 7zip file is provided, including the network traffic and the network flow statistics for each entity. Moreover, a relevant diagram is provided, illustrating the corresponding cyberattack. In particular, for each entity, a folder is given, including (a) the relevant pcap file, (b) Transmission Control Protocol (TCP) / Internet Protocol (IP) network flow statistics in a Common Separated Value (CSV) format and (c) IEC 60870-5-104 flow statistics in a CSV format. The TCP/IP network flow statistics were generated by CICFlowMeter<sup>7</sup>, while the IEC 60870-5-104 flow statistics were generated based on a Custom IEC 60870-5-104 Python Parser<sup>8</sup>, taking full advantage of Scapy<sup>9</sup>.

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<sup>7</sup> CICFlowMeter - <https://github.com/ahlashkari/CICFlowMeter>

<sup>8</sup> This parser is provided after a communication with the authors.

<sup>9</sup> Scapy - <https://scapy.net/>

### 3. Dataset Structure

The dataset consists of the following files:

- **20200425\_UOWM\_IEC104\_Dataset\_m\_sp\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the M\_SP\_NA\_1 attack.
- **20200426\_UOWM\_IEC104\_Dataset\_c\_ci\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the C\_CI\_NA\_1\_DoS attack.
- **20200426\_UOWM\_IEC104\_Dataset\_c\_ci\_na\_1.7z:** A 7zip file including the pcap and CSV files related to C\_CI\_NA\_1 attack.
- **20200427\_UOWM\_IEC104\_Dataset\_c\_se\_na\_1.7z:** A 7zip file including the pcap and CSV files related to the C\_SE\_NA\_1 attack.
- **20200428\_UOWM\_IEC104\_Dataset\_c\_sc\_na\_1.7z:** A 7zip file including the pcap and CSV files related to the C\_SC\_NA\_1 attack.
- **20200428\_UOWM\_IEC104\_Dataset\_c\_se\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the C\_SE\_NA\_1\_DoS attack.
- **20200429\_UOWM\_IEC104\_Dataset\_c\_sc\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the C\_SC\_NA\_1\_DoS attack.
- **20200605\_UOWM\_IEC104\_Dataset\_c\_rd\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the C\_RD\_NA\_1\_DoS attack.
- **20200605\_UOWM\_IEC104\_Dataset\_c\_rd\_na\_1.7z:** A 7zip file including the pcap and CSV files related to the C\_RD\_NA\_1 attack.
- **20200606\_UOWM\_IEC104\_Dataset\_c\_rp\_na\_1\_DoS.7z:** A 7zip file including the pcap and CSV files related to the C\_RP\_NA\_1\_DoS attack.
- **20200606\_UOWM\_IEC104\_Dataset\_c\_rp\_na\_1.7z:** A 7zip file including the pcap and CSV files related to the C\_RP\_NA\_1 attack.
- **20200608\_UOWM\_IEC104\_Dataset\_mitm\_drop.7z:** A 7zip file including the pcap and CSV files related to the MITM attack.
- **Balanced\_IEC104\_Train\_Test\_CSV\_Files.zip:** This zip file includes balanced CSV files from CICFlowMeter and the Custom IEC 60870-5-104 Python Parser that could be utilised for training ML and DL methods. The zip file includes different folders for the corresponding flow timeout values used for CICFlowMeter and IEC 60870-5-104 Python Parser, respectively.

Each 7zip file includes respective folders related to the entities/devices (described in the following section) participating in each attack. In particular, for each entity/device, there is a folder including (a) the overall network traffic (pcap file) related to this entity/device during each attack, (b) the TCP/IP network flow statistics (CSV file) from CICFlowMeter for the overall network traffic, (c) the IEC 60870-5-104 network traffic (pcap file) related to this entity/device during each attack, (d) the TCP/IP network flow statistics

(CSV file) from `CICFlowMeter` for the IEC 60870-5-104 network traffic, (e) the IEC 60870-5-104 flow statistics (CSV file) from the `Custom IEC 60870-5-104 Python Parser` for the IEC 60870-5-104 network traffic and finally, (f) an image showing how the attack was executed. Finally, it is noteworthy that the network flow from both `CICFlowMeter` and `Custom IEC 60870-5-104 Python Parser` in each CSV file are **labelled** based on the IEC 60870-5-104 cyberattacks executed for the generation of this dataset. The description of these attacks is given in the following section, while the various features from `CICFlowMeter` and `Custom IEC 60870-5-104 Python Parser` are presented in Section 5.

## 4. Testbed & IEC 60870-5-104 Attacks

Fig. 1. shows the testbed created for generating this dataset. It is composed of five virtual RTU devices emulated by IEC TestServer and two real RTU devices. Moreover, there is another workstation which plays the role of Master Terminal Unit (MTU) and HMI, sending legitimate IEC 60870-5-104 commands to the corresponding RTUs. For this purpose, the workstation uses QTester104. In addition, there are three attackers that act as malicious insiders executing the following cyberattacks against the aforementioned RTUs. Finally, the network traffic data of each entity/device was captured through tshark.

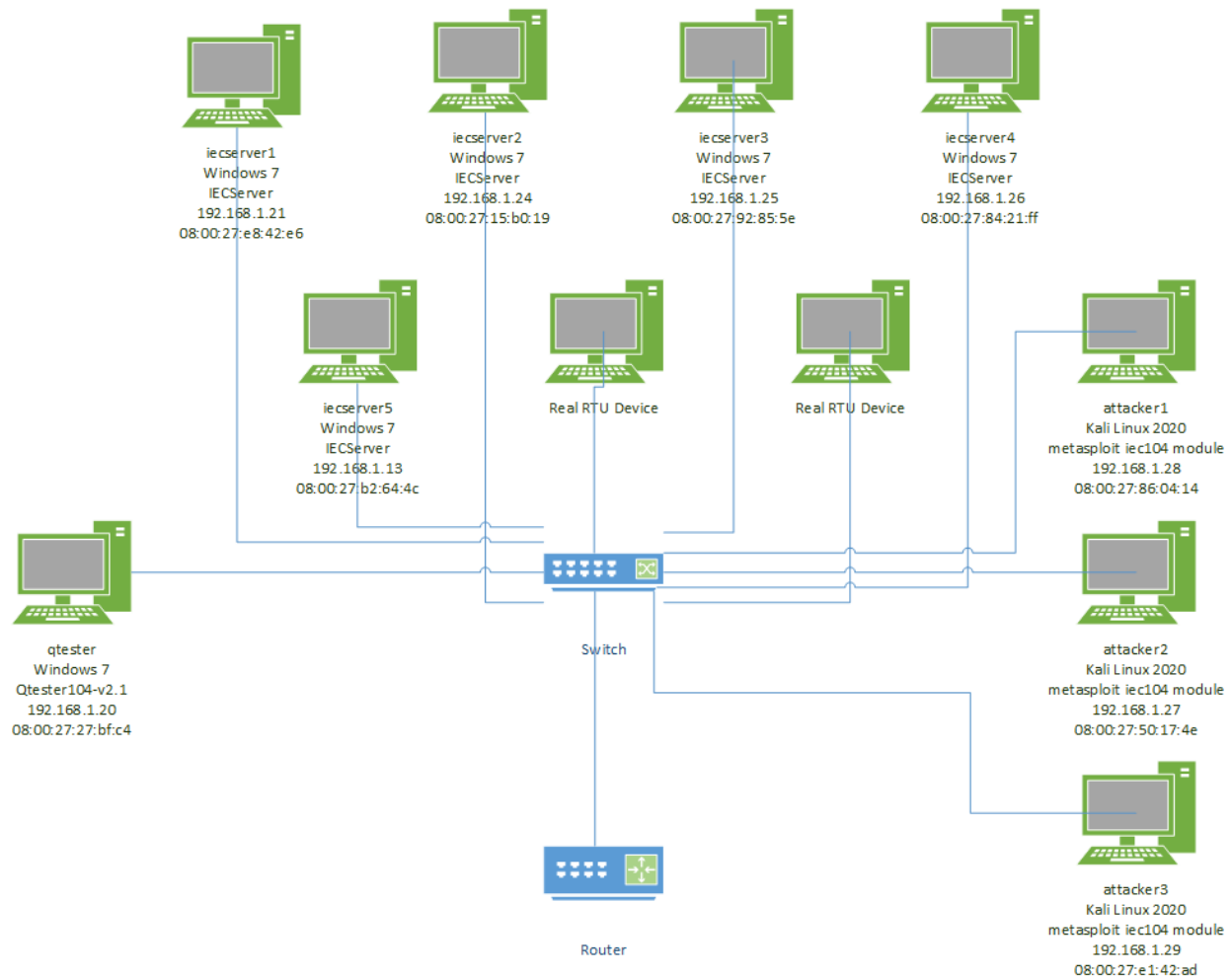


Fig. 1: Testbed used for the generation of the IEC 60870-5-104 Intrusion Detection Dataset

Table 1: IEC 60870-5-104 Cyberattacks Description

| IEC 60870-5-104<br>Cyberattack Description | Description                                                                                                                                                            | Dataset Files                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| MITM Drop                                  | During this attack, the cyberattacker is placed between two endpoints, thus monitoring and dropping the network traffic exchanged.                                     | 20200608_UOWM_IEC104_Dataset_mitm_drop.7z |
| C_CI_NA_1                                  | The C_CI_NA_1 is a Counter Interrogation command in the control direction. This cyberattack sends unauthorised IEC 60870-5-104 C_CI_NA_1 packets to the target system. | 20200426_UOWM_IEC104_Dataset_c_ci_na_1.7z |
| C_SC_NA_1                                  | The C_SC_NA_1 command is a single command. This cyberattack sends unauthorised C_SC_NA_1 60870-5-104 packets to the target system                                      | 20200428_UOWM_IEC104_Dataset_c_sc_na_1.7z |
| C_SE_NA_1                                  | The C_SE_NA_1 command is a set-point command with normalised values. This cyberattack sends unauthorised IEC 60870-5-104 C_SE_NA_1 packets to the target system.       | 20200427_UOWM_IEC104_Dataset_c_se_na_1.7z |
| C_RD_NA_1                                  | The C_RD_NA_1 command is a read command. This cyberattack sends unauthorised IEC 60870-5-104 C_RD_NA_1 packets to the target system.                                   | 20200605_UOWM_IEC104_Dataset_c_rd_na_1.7z |
| C_RP_NA_1                                  | The C_RP_NA_1 command is a reset command. This cyberattack sends                                                                                                       | 20200606_UOWM_IEC104_Dataset_c_rp_na_1.7z |



|               |                                                                              |                                               |
|---------------|------------------------------------------------------------------------------|-----------------------------------------------|
|               | unauthorised IEC 60870-5-104 C_RP_NA_1 packets to the target system.         |                                               |
| M_SP_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 M_SP_NA_1 packets. | 20200425_UOWM_IEC104_Dataset_m_sp_na_1_DoS.7z |
| C_CI_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 C_CI_NA_1 packets. | 20200426_UOWM_IEC104_Dataset_c_ci_na_1_DoS.7z |
| C_SE_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 C_SE_NA_1 packets. | 20200428_UOWM_IEC104_Dataset_c_se_na_1_DoS.7z |
| C_SC_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 C_SC_NA_1 packets. | 20200429_UOWM_IEC104_Dataset_c_sc_na_1_DoS.7z |
| C_RD_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 C_RD_NA_1 packets. | 20200605_UOWM_IEC104_Dataset_c_rd_na_1_DoS.7z |
| C_RP_NA_1_DoS | This attack floods the target system with IEC 60870-5-104 C_RP_NA_1 packets. | 20200606_UOWM_IEC104_Dataset_c_rp_na_1_DoS.7z |

## 5. Features

The TCP/IP network flow statistics generated by `CICFlowMeter` are summarised below. **It is worth mentioning that the TCP/IP network flows and their statistics generated by `CICFlowMeter` are labelled based on the IEC 60870-5-104 attacks described above, thus allowing the training of ML/DL models.**

Table 2: CICFlowMeter TCP/IP Network Flow Statistics - Features

| Feature          | Description                                                  |
|------------------|--------------------------------------------------------------|
| Flow ID          | ID of the flow                                               |
| Src IP           | Source IP address                                            |
| Src Port         | Source TCP/UDP port                                          |
| Dst IP           | Destination IP address                                       |
| Dst Port         | Destination TCP/UDP port                                     |
| Protocol         | The protocol related to the corresponding flow               |
| Timestamp        | Flow timestamp                                               |
| Flow Duration    | Duration of the flow in Microsecond                          |
| Tot Fwd Pkts     | Total packets in the forward direction                       |
| Tot Bwd Pkts     | Total packets in the backward direction                      |
| TotLen Fwd Pkts  | Total size of packets in forward direction                   |
| TotLen Bwd Pkts  | Total size of packets in backward direction                  |
| Fwd Pkt Len Max  | Maximum size of packet in forward direction                  |
| Fwd Pkt Len Min  | Minimum size of packet in forward direction                  |
| Fwd Pkt Len Mean | Mean size of packet in forward direction                     |
| Fwd Pkt Len Std  | Standard deviation size of packet in forward direction       |
| Bwd Pkt Len Max  | Maximum size of packet in backward direction                 |
| Bwd Pkt Len Min  | Minimum size of packet in backward direction                 |
| Bwd Pkt Len Mean | Mean size of packet in backward direction                    |
| Bwd Pkt Len Std  | Standard deviation size of packet in backward direction      |
| Flow Byts/s      | Number of flow bytes per second                              |
| Flow Pkts/s      | Number of flow packets per second                            |
| Flow IAT Mean    | Mean time between two packets sent in the flow               |
| Flow IAT Std     | Standard deviation time between two packets sent in the flow |
| Flow IAT Max     | Maximum time between two packets sent in the flow            |
| Flow IAT Min     | Minimum time between two packets sent in the flow            |
| Fwd IAT Tot      | Total time between two packets sent in the forward direction |

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| Fwd IAT Mean   | Mean time between two packets sent in the forward direction                                      |
| Fwd IAT Std    | Standard deviation time between two packets sent in the forward direction                        |
| Fwd IAT Max    | Maximum time between two packets sent in the forward direction                                   |
| Fwd IAT Min    | Minimum time between two packets sent in the forward direction                                   |
| Bwd IAT Tot    | Total time between two packets sent in the backward direction                                    |
| Bwd IAT Mean   | Mean time between two packets sent in the backward direction                                     |
| Bwd IAT Std    | Standard deviation time between two packets sent in the backward direction                       |
| Bwd IAT Max    | Maximum time between two packets sent in the backward direction                                  |
| Bwd IAT Min    | Minimum time between two packets sent in the backward direction                                  |
| Fwd PSH Flags  | Number of times the PSH flag was set in packets travelling in the forward direction (0 for UDP)  |
| Bwd PSH Flags  | Number of times the PSH flag was set in packets travelling in the backward direction (0 for UDP) |
| Fwd URG Flags  | Number of times the URG flag was set in packets travelling in the forward direction (0 for UDP)  |
| Bwd URG Flags  | Number of times the URG flag was set in packets travelling in the backward direction (0 for UDP) |
| Fwd Header Len | Total bytes used for headers in the forward direction                                            |
| Bwd Header Len | Total bytes used for headers in the backward direction                                           |
| Fwd Pkts/s     | Number of forward packets per second                                                             |
| Bwd Pkts/s     | Number of backward packets per second                                                            |
| Pkt Len Min    | Minimum length of a packet                                                                       |
| Pkt Len Max    | Maximum length of a packet                                                                       |
| Pkt Len Mean   | Mean length of a packet                                                                          |
| Pkt Len Std    | Standard deviation length of a packet                                                            |
| Pkt Len Var    | Variance length of a packet                                                                      |
| FIN Flag Cnt   | Number of packets with FIN                                                                       |
| SYN Flag Cnt   | Number of packets with SYN                                                                       |
| RST Flag Cnt   | Number of packets with RST                                                                       |
| PSH Flag Cnt   | Number of packets with PUSH                                                                      |
| ACK Flag Cnt   | Number of packets with ACK                                                                       |
| URG Flag Cnt   | Number of packets with URG                                                                       |
| CWE Flag Count | Number of packets with CWE                                                                       |
| ECE Flag Cnt   | Number of packets with ECE                                                                       |
| Down/Up Ratio  | Download and upload ratio                                                                        |

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Pkt Size Avg      | Average size of packet                                                             |
| Fwd Seg Size Avg  | Average size observed in the forward direction                                     |
| Bwd Seg Size Avg  | Average size observed in the backward direction                                    |
| Fwd Byts/b Avg    | Average number of bytes bulk rate in the forward direction                         |
| Fwd Pkts/b Avg    | Average number of packets bulk rate in the forward direction                       |
| Fwd Blk Rate Avg  | Average number of bulk rate in the forward direction                               |
| Bwd Byts/b Avg    | Average number of bytes bulk rate in the backward direction                        |
| Bwd Pkts/b Avg    | Average number of packets bulk rate in the backward direction                      |
| Bwd Blk Rate Avg  | Average number of bulk rate in the backward direction                              |
| Subflow Fwd Pkts  | The average number of packets in a sub flow in the forward direction               |
| Subflow Fwd Byts  | The average number of bytes in a sub flow in the forward direction                 |
| Subflow Bwd Pkts  | The average number of packets in a sub flow in the backward direction              |
| Subflow Bwd Byts  | The average number of bytes in a sub flow in the backward direction                |
| Init Fwd Win Byts | The total number of bytes sent in initial window in the forward direction          |
| Init Bwd Win Byts | The total number of bytes sent in initial window in the backward direction         |
| Fwd Act Data Pkts | Count of packets with at least 1 byte of TCP data payload in the forward direction |
| Fwd Seg Size Min  | Minimum segment size observed in the forward direction                             |
| Active Mean       | Mean time a flow was active before becoming idle                                   |
| Active Std        | Standard deviation time a flow was active before becoming idle                     |
| Active Max        | Maximum time a flow was active before becoming idle                                |
| Active Min        | Minimum time a flow was active before becoming idle                                |
| Idle Mean         | Mean time a flow was idle before becoming active                                   |
| Idle Std          | Standard deviation time a flow was idle before becoming active                     |
| Idle Max          | Maximum time a flow was idle before becoming active                                |
| Idle Min          | Minimum time a flow was idle before becoming active                                |
| Label             | Attack label                                                                       |

The IEC 60870-5-104 flow statistics generated by `IEC 60870-5-104 Python Parser` are summarised below. **It is worth mentioning that the IEC 60870-5-104 flows and their statistics generated by `IEC 60870-5-104 Python Parser` are labelled based on the IEC 60870-5-104 attacks described above, thus allowing the training of ML/DL models.**

**Table 3: IEC 60870-5-104 Flow Statistics – Features**

| Feature  | Field description                                            |
|----------|--------------------------------------------------------------|
| flow id  | ID of the flow                                               |
| protocol | The relevant protocol of the flow. It equals IEC 60870-5-104 |

-> for isolation  
->

|                           |                                                                                  |
|---------------------------|----------------------------------------------------------------------------------|
| src ip                    | The source IP address of the flow. It is defined with the first relevant packet. |
| dst ip                    | The destination IP address of the flow.                                          |
| src port                  | The source TCP/UDP port.                                                         |
| dst port                  | The destination TCP/UDP port.                                                    |
| flow idle time max        | The maximum time where the flow was idle                                         |
| flow idle time min        | The minimum time where the flow was idle                                         |
| flow idle time mean       | The time mean where the flow was idle                                            |
| flow idle time std        | The time standard deviation where the flow was idle                              |
| flow idle time variance   | The time variance where the flow was idle                                        |
| flow active time max      | The maximum time where the flow was active                                       |
| flow active time min      | The minimum time where the flow was active                                       |
| flow active time mean     | The time mean where the flow was active                                          |
| flow active time std      | The time standard deviation where the flow was active                            |
| flow active time variance | The time variance where the flow was active                                      |
| flow IAT max              | The maximum interarrival time                                                    |
| fw IAT max                | The maximum interarrival time in the forward direction                           |
| bw IAT max                | The maximum interarrival time in the backyard direction                          |
| flow IAT min              | The minimum interarrival time                                                    |
| fw IAT min                | The minimum interarrival time in the forward direction                           |
| bw IAT min                | The minimum interarrival time in the backyard direction                          |
| flow IAT mean             | The mean of the interarrival time                                                |
| fw IAT mean               | The mean of the interarrival time in the forward direction                       |
| bw IAT mean               | The mean of the interarrival time in the backyard direction                      |
| flow IAT std              | The standard deviation of the inter arrival time                                 |
| fw IAT std                | The standard deviation of the inter arrival time in the forward direction        |
| bw IAT std                | The standard deviation of the inter arrival time in the backyard direction       |
| flow IAT tot              | The total number of the interarrival times                                       |
| fw iAT tot                | The total number of the interarrival times in the forward direction              |
| bw IAT tot                | The total number of the interarrival times in the backyard direction             |
| flow iec104 packets/s     | The number of IEC 60870-51-04 packets per second                                 |
| fw iec104 packets/s       | The number of IEC 60870-51-04 packets per second in the forward direction        |

|                                            |                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| bw iec104 packets/s                        | The number of IEC 60870-51-04 packets per second in the backyard direction                                                          |
| flow iec104 bytes/s                        | The sum of APDU lengths per second                                                                                                  |
| fw iec104 bytes/s                          | The sum of APDU lengths per second in the forward direction                                                                         |
| bw iec104 bytes/s                          | The sum of APDU lengths per second in the backyard direction                                                                        |
| flow packet APDU length max                | The maximum value of the APDU lengths                                                                                               |
| flow packet APDU length min                | The minimum value of the APDU lengths                                                                                               |
| flow packet APDU length mean               | Mean of the APDU lengths                                                                                                            |
| flow packet APDU length std                | The standard deviation of the APDU lengths                                                                                          |
| flow packet APDU length var                | Variance of the APDU lengths                                                                                                        |
| fw packet APDU length max                  | The maximum value of the APDU lengths in the forward direction                                                                      |
| fw packet APDU length min                  | The minimum value of the APDU lengths in the forward direction                                                                      |
| fw packet APDU length mean                 | Mean of the APDU lengths in the forward direction                                                                                   |
| fw packet APDU length std                  | The standard deviation of the APDU lengths in the forward direction                                                                 |
| fw packet APDU length var                  | The variance of the APDU lengths in the forward direction                                                                           |
| bw packet APDU length max                  | The maximum value of the APDU lengths in the backyard direction                                                                     |
| bw packet APDU length min                  | The minimum value of the APDU lengths in the backyard direction                                                                     |
| bw packet APDU length mean                 | Mean of the APDU lengths in the backyard direction                                                                                  |
| bw packet APDU length std                  | The standard deviation of the APDU lengths in the backyard direction                                                                |
| bw packet APDU length var                  | The variance of the APDU lengths in the backyard direction                                                                          |
| total flow packets                         | Total flow packets                                                                                                                  |
| total fw packets                           | Total flow packets in the forward direction                                                                                         |
| total bw packets                           | Total flow packets in the backyard direction                                                                                        |
| flow packets APDU total length             | The sum of all APDU lengths                                                                                                         |
| fw packets APDU total length               | The sum of all APDU lengths in the forward direction                                                                                |
| bw packets APDU total length               | The sum of all APDU lengths in the backyard direction                                                                               |
| flow duration                              | Flow duration in seconds                                                                                                            |
| flow down/up ratio                         | The fraction between the IEC 60870-5-104 packets in the backyard direction and the IEC 60870-5-104 packets in the forward direction |
| flow total IEC104_I_Message_SeqIOA packets | The total number of the I-format APCI packets that have more than one information objects                                           |
| fw total IEC104_I_Message_SeqIOA packets   | The total number of the I-format APCI packets that have more than one information objects in the forward direction                  |
| bw total IEC104_I_Message_SeqIOA packets   | The total number of the I-format APCI packets that have more than one information objects in the backyard direction                 |

|                                               |                                                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| flow total IEC104_I_Message_SingleIOA packets | The total number of the I-format APCI packets that have one information object in ASDU                           |
| fw total IEC104_I_Message_SingleIOA packets   | The total number of the I-format APCI packets that have one information object in ASDU in the forward direction  |
| bw total IEC104_I_Message_SingleIOA packets   | The total number of the I-format APCI packets that have one information object in ASDU in the backyard direction |
| flow total IEC104_S_Message packets           | The total number of the S-format APCI packets                                                                    |
| fw total IEC104_S_Message packets             | The total number of the S-format APCI packets in the forward direction                                           |
| bw total IEC104_S_Message packets             | The total number of the S-format APCI packets in the backyard direction                                          |
| flow total IEC104_U_Message packets           | The total number of the U-format APCI packets                                                                    |
| fw total IEC104_U_Message packets             | The total number of the U-format APCI packets in the forward direction                                           |
| bw total IEC104_U_Message packets             | The total number of the U-format APCI packets in the backyard direction                                          |
| fw URG flag amount                            | The number of the URG flags in the forward direction                                                             |
| fw PSH flag amount                            | The number of the PSH flags in the forward direction                                                             |
| bw URG flag amount                            | The number of the URG flags in the backyard direction                                                            |
| bw PSH flag amount                            | The number of the PSH flags in the backyard direction                                                            |
| flow SYN flag count                           | The number of the TCP SYN packets                                                                                |
| flow RST flag count                           | The number of the TCP RST packets                                                                                |
| flow PSH flag count                           | The number of the TCP PSH packets                                                                                |
| flow ACK flag count                           | The number of the TCP ACK packets                                                                                |
| flow URG flag count                           | The number of the TCP URG packets                                                                                |
| flow CWE flag count                           | The number of the TCP CWE packets                                                                                |
| flow ECE flag count                           | The number of the TCP ECE packets                                                                                |
| fw_subflow_packets                            | The average number of packets in a sub flow in the forward direction                                             |
| bw_subflow_packets                            | The average number of packets in a sub flow in the backward direction                                            |
| fw_subflow_bytes                              | The average number of bytes in a sub flow in the forward direction                                               |
| bw_subflow_bytes                              | The average number of bytes in a sub flow in the backward direction                                              |
| flow start timestamp                          | The timestamp of the flow. It is defined with the first relevant packet.                                         |
| fw avg bytes/bulk                             | Average number of bytes bulk rate in the forward direction                                                       |
| bw avg bytes/bulk                             | Average number of bytes bulk rate in the backyard direction                                                      |
| fw avg bulk rate                              | Average number of bulk rate in the forward direction                                                             |

|                                                      |                                                                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| bw avg bulk rate                                     | Average number of bulk rate in the backyard direction                                                  |
| fw avg packets/bulk                                  | Average number of packets bulk rate in the forward direction                                           |
| bw avg packets/bulk                                  | Average number of packets bulk rate in the backyard direction                                          |
| init fw window bytes                                 | The window size of the first packet in the forward direction                                           |
| init bw window bytes                                 | The window size of the first packet in the backyard direction                                          |
| fw TCP total header length                           | The length of the TCP headers in the forward direction                                                 |
| bw TCP total header length                           | The length of the TCP headers in the backyard direction                                                |
| cot=1                                                | The total number of the IEC 60870-5-104 packets where COT = 1 (periodic,cyclic)                        |
| cot=2                                                | The total number of the IEC 60870-5-104 packets where COT = 2 (background interrogation)               |
| cot=3                                                | The total number of the IEC 60870-5-104 packets where COT = 3 (spontaneous)                            |
| cot=4                                                | The total number of the IEC 60870-5-104 packets where COT = 4 (initialized)                            |
| cot=5                                                | The total number of the IEC 60870-5-104 packets where COT = 5 (interrogation)                          |
| cot=6                                                | The total number of the IEC 60870-5-104 packets where COT = 6 (activation)                             |
| cot=7                                                | The total number of the IEC 60870-5-104 packets where COT = 7 (confirmation activation)                |
| cot=8                                                | The total number of the IEC 60870-5-104 packets where COT = 8 (deactivation)                           |
| cot=9                                                | The total number of the IEC 60870-5-104 packets where COT = 9 (confirmation deactivation)              |
| cot=10                                               | The total number of the IEC 60870-5-104 packets where COT = 10 (termination activation)                |
| cot=11                                               | The total number of the IEC 60870-5-104 packets where COT = 11 (feedback, caused by distant command)   |
| cot=12                                               | The total number of the IEC 60870-5-104 packets where COT = 12 (feedback, caused by local command)     |
| cot=13                                               | The total number of the IEC 60870-5-104 packets where COT = 13 (COT data transmission)                 |
| cot=20                                               | The total number of the IEC 60870-5-104 packets where COT = 20 (interrogated by general interrogation) |
| type_id_process_information_in_monit<br>or_direction | The total number of the IEC 60870-5-104 packets where TypeID is in the range 1-40                      |
| type_id_process_information_in_contro<br>l_direction | The total number of the IEC 60870-5-104 packets where TypeID is in the range 45-51                     |



|                                                 |                                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------|
| type_id_system_information_in_monitor_direction | The total number of the IEC 60870-5-104 packets where TypeID is in the range 70      |
| type_id_system_information_in_control_direction | The total number of the IEC 60870-5-104 packets where TypeID is in the range 100-106 |
| type_id_parameter_in_control_direction          | The total number of the IEC 60870-5-104 packets where TypeID is in the range 110-113 |
| type_id_file_transfer                           | The total number of the IEC 60870-5-104 packets where TypeID is in the range 120-126 |
| Label                                           | Attack label                                                                         |

## 6. Citation

Please cite the following paper when using this dataset:

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## References

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- [2] A. Gharib, I. Sharafaldin, A. H. Lashkari and A. A. Ghorbani, "An Evaluation Framework for Intrusion Detection Dataset," 2016 International Conference on Information Science and Security (ICISS), 2016, pp. 1-6, doi: 10.1109/ICISSEC.2016.7885840.